



FINAL YEAR ENGINEERING PROJECT

Smart Medicine Availability & Intelligent Janaushadhi Recommendation System

PAGE GUIDE

Project	Smart Medicine Availability & Intelligent Janaushadhi Recommendation System
Document	PAGE GUIDE
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Every Page Explained — Purpose, User Flow, Components, and Backend APIs

Public Pages

HomePage (`/`)

Purpose: The public landing page that explains the project to first-time visitors. Encourages users to register or log in.

Layout: `MainLayout` (Navbar + Footer)

Sections (lazy-loaded subsections):

1. **HeroSection** — Full-width banner with search CTA and headline
2. **AboutSection** — What the system does
3. **JanAushadhiSection** — What Jan Aushadhi is and why it matters
4. **FeaturesSection** — Key features of the app
5. **HowItWorksSection** — Step-by-step user journey
6. **BenefitsSection** — Benefits of using generic medicines
7. **StatsSection** — Impact numbers (pharmacies, savings, medicines)
8. **FaqSection** — Frequently asked questions
9. **CtaSection** — Final call-to-action with register button

Components Used: Navbar, Footer, Container, SectionHeader, Button, Badge

User Flow:

```

Visitor arrives at homepage
  → Scrolls through sections
  → Clicks "Get Started" or "Search Medicines"
  → Redirected to Login (if not authenticated)
  
```

Future Backend APIs:

```

GET /api/v1/stats/summary → StatsSection (total pharmacies, medicines, users)
  
```

Authentication Pages

Layout: `AuthLayout` (centered card, no sidebar)

LoginPage (`/login`)

Purpose: Authenticates existing users.

Features:

- Email + Password form
- "Remember Me" checkbox

- Links to Register and Forgot Password
- Role-based redirect after login (patient → /dashboard, admin → /admin/dashboard)

Form Validation (Zod `loginSchema`):

- Email: required, valid email format
- Password: required (min 1 character)

Components Used: Input, PasswordInput, Checkbox, Button, Divider

User Flow:

```
User enters email + password
  → Clicks "Login"
  → AuthContext.login() called
  → On success: redirect to role dashboard
  → On failure: show error message
```

Future Backend APIs:

```
POST /api/v1/auth/login
Request: { email, password }
Response: { user: {...}, accessToken, refreshToken }
```

RegisterPage (`/register`)

Purpose: Creates a new patient account.

Features:

- Full name, email, mobile, password, confirm password
- Password strength indicator
- Terms & Conditions checkbox
- Submit → navigate to OTP verification

Form Validation (Zod `registerSchema`):

- Full name: 2–60 characters, letters only
- Email: valid format
- Mobile: valid Indian 10-digit number
- Password: 8+ chars, uppercase, lowercase, digit
- Passwords match
- Must accept terms

Components Used: Input, PasswordInput, PasswordStrength, Checkbox, Button

Future Backend APIs:

```
POST /api/v1/auth/register
Request: { fullName, email, mobile, password }
Response: { message: "OTP sent to email" }
```

VerifyOtpPage (`/verify-otp`)

Purpose: Verifies the 6-digit OTP sent to the user's email after registration or forgot password.

Features:

- 6-box OTP input with auto-focus advance

- 60-second countdown timer with "Resend OTP" button
- Shows the email address the OTP was sent to (passed via navigation state)

Components Used: `OtpInput`, `Button`, `Spinner`

Future Backend APIs:

```
POST /api/v1/auth/verify-otp
Request: { email, otp }
Response: { message: "Account verified" }

POST /api/v1/auth/resend-otp
Request: { email }
Response: { message: "OTP resent" }
```

ForgotPasswordPage (`/forgot-password`)

Purpose: Initiates the password reset flow by sending an OTP to the user's email.

Components Used: `Input`, `Button`

Future Backend APIs:

```
POST /api/v1/auth/forgot-password
Request: { email }
Response: { message: "Reset OTP sent" }
```

ResetPasswordPage (`/reset-password`)

Purpose: Allows the user to set a new password after OTP verification.

Components Used: `PasswordInput`, `PasswordStrength`, `Button`

Future Backend APIs:

```
POST /api/v1/auth/reset-password
Request: { email, otp, newPassword }
Response: { message: "Password reset successfully" }
```

Patient / User Pages

Layout: `UserLayout` (Sidebar + TopBar)

Route Guard: `ProtectedRoute` — roles: `patient`, `doctor`

UserDashboard (`/dashboard`)

Purpose: The patient's home screen after login. Shows a personalised overview of their activity and quick-access shortcuts.

Sections:

1. Welcome banner with user name
2. Quick action buttons (Search, Nearby Pharmacy, Notifications)

3. Recent searches
4. Saved medicines
5. Recent notifications (last 3)
6. Health tips section
7. Jan Aushadhi awareness banner
8. Stats (searches made, medicines saved)
9. Profile completion indicator
10. Nearby pharmacy teaser

Components Used: InfoCard, NotificationCard, MedicineCard, Avatar, Badge, Button, Toggle

Future Backend APIs:

GET /api/v1/users/me/dashboard → Dashboard summary data
 GET /api/v1/users/me/saved-medicines → Saved medicines list
 GET /api/v1/users/me/notifications?limit=3 → Recent notifications

MedicineSearchPage (`/search`)

Purpose: The main medicine search interface.

Features:

- Full-width search bar with autocomplete suggestions
- Quick action shortcuts (Search by Composition, Scan Barcode, Voice Search)
- Recent searches (stored in localStorage)
- Popular medicine suggestions
- Medicine category browse grid
- Advanced filter panel (type, availability, price range, manufacturer)
- Loading, empty, and error states

Components Used: SearchBar, Button, Badge, EmptyState, Skeleton, useDebounce

Future Backend APIs:

GET /api/v1/medicines/search?q={query} → Search suggestions
 GET /api/v1/medicines/popular → Popular medicines
 GET /api/v1/medicines/categories → Category list

SearchResultsPage (`/search/results`)

Purpose: Shows the list of medicines matching the search query.

Features:

- Results count and query display
- Filter sidebar (type, price range, availability, manufacturer, category)
- Sort options (relevance, price low/high, name A-Z)
- Grid / List view toggle
- Compare mode — select up to 4 medicines
- CompareBar — sticky bottom bar showing selected medicines
- Pagination

Components Used: SearchResultCard, Badge, Pagination, EmptyState, Skeleton, Button

Future Backend APIs:

```
GET /api/v1/medicines/search?q={query}&type=generic&sort=price_asc&page=1&limit=10
Response: { medicines: [...], total, page, totalPages }
```

MedicineDetailsPage (`/medicine/:id`)

Purpose: Full detail view of a single medicine.

Sections:

1. Medicine header (name, manufacturer, price, availability badges)
2. Action panel (View Generic, Add to Saved, Share, Nearby Pharmacies)
3. MedicineInfoTabs: Overview | Composition | Usage | Side Effects | Storage
4. Generic Recommendation section (teaser with link to full page)
5. Nearby pharmacies teaser
6. Price comparison section
7. Related medicines
8. Disclaimer

Components Used: Badge, Button, Breadcrumb, Modal, MedicineCard, PharmacyCard

Future Backend APIs:

```
GET /api/v1/medicines/:id
Response: { id, name, genericName, composition, manufacturer, price, mrp, ... }
```

```
GET /api/v1/medicines/alternatives/:id
Response: [{ id, name, price, isJanAushadhi, ... }]
```

GenericRecommendationPage (`/medicine/:id/generic`)

Purpose: A full-page view of the Jan Aushadhi generic alternatives for a specific branded medicine. This is the core feature of the system.

Sections:

1. Branded medicine summary
2. Why generics section (educational)
3. Generic alternatives list with price comparison
4. Jan Aushadhi pharmacy locator teaser
5. Cost savings calculator
6. Doctor consultation note
7. Savings summary
8. Evidence section
9. How to switch guide
10. Awareness banner
11. Legal disclaimer
12. FAQ section
13. Share this recommendation button
14. Call to action (View Nearby Pharmacies)
15. Health disclaimer footer

Components Used: MedicineCard, Badge, Button, Modal, InfoCard

Future Backend APIs:

```
GET /api/v1/medicines/:id/generic-recommendation
Response: {
  branded: {...},
  generics: [...],
  savings: { amount, percentage }
}
```

NearbyPharmaciesPage (`/pharmacies/nearby`)

Purpose: Shows Jan Aushadhi pharmacies near the user on an interactive map.

Features:

- React Leaflet interactive map (OpenStreetMap tiles)
- Pharmacy marker pins
- Click marker → show pharmacy info popup
- Pharmacy list panel (sortable by distance)
- Filter panel (open now, has this medicine)
- Get Directions button (opens Google Maps)
- Search radius control

Components Used: PharmacyCard, Button, Badge, Spinner, EmptyState

Future Backend APIs:

```
GET /api/v1/pharmacies/nearby?lat={lat}&lng={lng}&radius={km}
Response: [{
  id, name, address, lat, lng, distance, isOpen, phone
}]
```

NotificationsPage (`/notifications`)

Purpose: Shows all notifications for the logged-in user.

Features:

- Tab filter: All / Unread / Medicines / Alerts
- Mark all as read button
- Individual mark-read and dismiss per notification

Components Used: NotificationCard, Badge, Button, EmptyState

Future Backend APIs:

```
GET /api/v1/users/me/notifications
PATCH /api/v1/users/me/notifications/:id/read
DELETE /api/v1/users/me/notifications/:id
```

ProfilePage (`/profile`)

Purpose: Allows the user to view and edit their profile information.

Sections:

- Avatar with upload option
- Personal info (name, email, phone)
- Change password
- Notification preferences (Toggle)
- Delete account

Components Used: Avatar, Input, PasswordInput, Toggle, Button

Future Backend APIs:

```
GET    /api/v1/users/me
PUT    /api/v1/users/me
DELETE /api/v1/users/me
```

Pharmacy Pages

Layout: [PharmacyLayout](#) | **Guard:** PHARMACIST role

PharmacyDashboard (`/pharmacy/dashboard`)

Purpose: Overview dashboard for pharmacists showing inventory summary, recent orders, and quick actions.

Components Used: InfoCard, Badge, Button, Avatar

Future Backend APIs:

```
GET /api/v1/pharmacy/dashboard
GET /api/v1/pharmacy/stats
```

InventoryPage (`/pharmacy/inventory`)

Purpose: Full inventory management — searchable, sortable, paginated medicine stock table.

Features:

- Search by name or composition
- Sort by name, stock, expiry
- Filter by category, availability
- Add New button → MedicineFormPage
- Edit button per row → MedicineFormPage in edit mode
- Delete with ConfirmDialog
- Pagination
- Low stock warning badges

Components Used: Button, Badge, Pagination, ConfirmDialog, Skeleton, EmptyState, SearchBar

Future Backend APIs:

```
GET    /api/v1/inventory?search={q}&page=1&limit=10
POST   /api/v1/inventory
PUT    /api/v1/inventory/:id
DELETE /api/v1/inventory/:id
```

MedicineFormPage (`/pharmacy/inventory/add` and `/pharmacy/inventory/edit/:id`)

Purpose: Add or edit a medicine stock entry. Same form used for both operations — the presence of `:id` in the URL determines edit mode.

Components Used: Input, Select, Textarea, Button, FormField

Future Backend APIs:

```
POST /api/v1/inventory      (add)
PUT  /api/v1/inventory/:id  (edit)
GET  /api/v1/inventory/:id  (load for edit)
```

Admin Pages

Layout: [AdminLayout](#) | **Guard:** `ADMIN` role

All admin pages use a shared [AdminTable](#) component and mock data from [pages/admin/data/](#).

Page	Route	Purpose
AdminDashboard	/admin/dashboard	System overview with key metrics
AdminUsers	/admin/users	User management table
AdminPharmacies	/admin/pharmacies	Pharmacy management table
AdminMedicines	/admin/medicines	Medicine catalogue management
AdminGenericMapping	/admin/generic-mapping	Map branded → generic medicines
AdminAnalytics	/admin/analytics	Usage analytics and charts
AdminReports	/admin/reports	Generate and download reports
AdminNotifications	/admin/notifications	Send system notifications
AdminActivity	/admin/activity	System activity and audit logs
AdminRoles	/admin/roles	Role and permission management
AdminSettings	/admin/settings	System configuration

Error Pages

Page	Route	Purpose
NotFoundPage	/*	404 — page not found
UnauthorizedPage	/unauthorized	403 — access denied
SessionExpiredPage	/session-expired	JWT expired
ServerErrorPage	(AppErrorBoundary)	Caught React crash
OfflinePage	(App.jsx)	Browser is offline